

Designing N-of-1 Trials for a Biomarker-Treatment Interaction: What Drives Power, and How to Choose a Design

2026-10-02 18:12 PDT

Author. pmsimstats team

1. Summary

docs/37 derived closed-form moments and power for the paired-difference interaction statistic (E9) in N-of-1 and crossover designs. This paper uses those expressions to ask which design characteristics determine power, and to search the design space for the best design. Without carryover the interaction slope is the same in every design, so a design affects power only through the noise of each patient's on-minus-off contrast and, under carryover, through how much of the biomarker's moderation survives at the off-drug visits. Every result below follows from that, and every number is a closed-form evaluation except where Monte Carlo is stated.

1. **What matters depends on how within-patient correlation behaves.** Under compound symmetry (CS), the published construct, correlation does not decay with time, so the arrangement of on and off visits is irrelevant: only the balance of on and off visits and the blinding matter. Under AR(1) correlation, the contrast cancels noise only between visits that are close in time, so interleaving on and off visits closely dominates (Sections 3 and 4).
2. **The Hybrid design's advantage is interleaving, and its open-label lead-in costs power.** Under CS, Hybrid is not the most powerful design: blinded balanced designs, including the crossover, match or beat it, and running the Hybrid schedule blinded raises its power from 0.72 to 0.80. Under AR(1), Hybrid beats the crossover because its blinded-discontinuation visits sit next to on-drug visits and it switches three or four times; designs that alternate more often do better still (Section 3).
3. **More on-off switches help only under AR(1), and carryover limits them.** Under CS more switches never help and, with carryover, hurt: at a two-week response half-life the best one-switch design has power 0.731 and the best seven-switch design 0.559. Under AR(1) with 20% measurement error, the best seven-switch design has 0.435 at a one-week half-life against 0.221 for one switch (Section 4).
4. **There is an optimal time between visits under AR(1), roughly two to six response half-lives.** Under CS wider spacing is always better, up to a plateau reached when off-drug visits fall about two to five response half-lives after discontinuation, depending on the design. Under

AR(1) the optimum is interior: close spacing cancels shared noise, wide spacing lets carryover wash out. For an alternating design it is about twice the half-life without measurement error and four to six times it when half the noise is measurement error (Section 5).

5. **The exhaustive search confirms the principles.** Over all 119 counterbalanced two-sequence designs with eight blinded visits, seven spacings, three correlation structures and three response half-lives, the best design is the crossover at the widest spacing under CS and the alternating design at two to four weeks under AR(1). A near-alternating design at four-week spacing never falls below 78% of the best attainable power in any of the nine conditions (Section 6).
6. **Open-label visits cost power when they replace blinded ones.** Each open-label lead-in visit that takes the place of a blinded visit lowers power, under every structure. Whether the open-label visits should enter the on/off contrast depends on the correlation structure (Section 7).
7. **The closed form is confirmed by simulation.** Direct Monte Carlo of 41 designs, including the best design in every condition, agrees with the closed form within two standard errors in every case (Section 8).

2. Framework

Criterion. For a design with contrast weights a_t ($1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug), `docs/37` shows that power is governed by the per-patient squared correlation between the contrast and the biomarker,

$$\rho^2 = \frac{(\gamma\sigma_B)^2}{\text{Var}(\Delta)}, \quad \gamma = -c_{bm} \frac{\sigma_{BR}}{\sigma_B} (1 - \bar{h}_{\text{off}}), \quad \text{Var}(\Delta) = a^\top C a,$$

where $\bar{h}_{\text{off}}$ is the average share of the biomarker's moderation still present at the off-drug visits and C the response covariance. The interaction itself, $c_{bm}\sigma_{BR}/\sigma_B$, does not depend on the design. A design therefore matters through two quantities only: the noise $a^\top C a$ of the contrast, and the surviving moderation $\bar{h}_{\text{off}}$. The *information* reported below is the patient-weighted ρ^2 .

Data-generating rule. Decayed mean moderation (`docs/37`, Section 5.6): the biomarker shifts the drug response by $c_{bm}\sigma_{BR}z h_t$, with $h_t = 1$ on drug and $2^{-t_{sd}/t_{1/2}}$ off drug, t_{sd} the time since discontinuation and $t_{1/2}$ the response half-life (`docs/38`). It has the same slope as graded coupling but no ceiling on c_{bm} , so the comparison of designs is not distorted by which effect sizes each design can represent. Under the covariance constructs, high-information designs have lower ceilings (`docs/37`, Section 5.8), an artifact this choice avoids.

Correlation structures. Three, from `docs/37` Section 5.8 and an extension of it:

- **CS**, the published construct: correlation 0.8 between any two visits of a factor.
- **AR(1) with measurement error**: the separable form $K \otimes A'$ with time kernel $A'_{ts} = (1 - \nu)\rho^{|w_t - w_s|} + \nu \mathbf{1}[t = s]$, $\rho = 0.8$ per week of separation, and a share ν of each factor's variance that is white measurement error, $\nu = 0.2$ or 0.5 .

The measurement-error share matters. Pure AR(1) has none, so visits placed very close together become nearly perfectly correlated, the contrast's noise vanishes, and the most powerful design would place visits as close together as possible for reasons that are not pharmacological. Symptom ratings carry measurement error, which sets a floor on the noise of any contrast; CS has an equivalent floor in its occasion-level term.

Designs and analysis. Eight post-baseline visits, $N = 70$, two counterbalanced sequences of 35 patients (an on/off pattern and its complement), blinded (expectancy 0.5) except where Section 7 adds an open-label lead-in, $c_{bm} = 0.25$. The analysis is E9 with sequence-specific intercepts, so that sequence and period differences, which E9's common intercept counts as noise (docs/37, Section 5.7), do not enter. Response half-lives of 0.5, 1 and 2 weeks.

3. What drives the noise of the contrast

Under CS, expanding $a^\top Ca$ (docs/37, Section 4.1) shows four contributions. For E9 the weights sum to zero and the person-level component cancels, leaving (per path, verified):

Design	Path	On / off visits	Balance term	Expectancy confound	Total Var(Δ)
Hybrid (published)	1, 2	5 / 3	17.5	3.2	44.1
	3, 4	4 / 4	16.4	5.0	45.0
Hybrid schedule, blinded	1, 2	5 / 3	17.5	0	38.3
	3, 4	4 / 4	16.4	0	35.9
OL+BDC	1	6 / 2	21.9	8.9	60.4
	2	5 / 3	17.5	12.8	56.4
CO, ABAB, alternating (blinded, 4 / 4)	all	4 / 4	16.4	0	35.9

Design	Path	On / off visits	Balance term	Expectancy confound	Total Var(Δ)
CO with an open-label first period	both	4 / 4	16.4	20.0	64.2

- **Balance.** The balance term is proportional to $1/n_{\text{on}} + 1/n_{\text{off}}$, smallest at four and four; five and three costs 7%, six and two 33%.
- **Expectancy confounding.** Open-label visits are always on drug, so they make the on/off contrast partly an open-label against blinded contrast ($\sum_t a_t e_t \neq 0$). The person-level part of the expectancy factor then no longer cancels and enters the contrast, together with that factor's doubled standard deviation at open-label visits.
- **Arrangement does not matter.** Every blinded four-and-four design has the same variance, 35.9, however its on and off visits are ordered.

So under CS the Hybrid design's open-label lead-in costs power: blinded, the same schedule has power 0.80 instead of 0.72 (E9), and giving the crossover an open-label first period lowers its power from 0.82 to 0.56.

Under AR(1), with constant expectancy the separable variance is a product, $\text{Var}(\Delta) = (s^\top K s)(a^\top A a)$, and the time part $a^\top A a$ is where designs differ. It is small when every on-drug visit has an off-drug visit close to it (per path, no measurement error, verified):

Design	Switches	$a^\top A a$	Var(Δ)
CO	1-2	0.828	212.8
OL+BDC	2	0.649-0.664	183.9-192.7
Hybrid	3-4	0.361-0.449	107.3-126.8
ABAB, every 2.5 weeks	3-4	0.352	90.6
ABAB at the Hybrid visit times	3-4	0.299	76.7
Alternating every visit, every 2.5 weeks	7-8	0.165	42.3

This is the Hybrid design's real advantage over the crossover and OL+BDC designs: its weekly blinded-discontinuation visits sit next to on-drug visits, and it switches three or four times rather than once or twice. The same idea taken further, an alternating design, has a quarter of the crossover's contrast noise.

4. More on-off switches

From the exhaustive search of Section 6, the best power attainable with a given number of switches in the first sequence, at the best spacing (verified):

Structure	$t_{1/2}$	1 switch	2	3	4	5	6	7
CS	0.5	0.765	0.765	0.764	0.764	0.763	0.763	0.763
CS	1	0.759	0.752	0.746	0.739	0.733	0.726	0.719
CS	2	0.731	0.703	0.675	0.645	0.616	0.584	0.559
AR(1), 20% error	0.5	0.268	0.386	0.453	0.512	0.510	0.522	0.547
AR(1), 20% error	1	0.221	0.282	0.339	0.389	0.397	0.404	0.435
AR(1), 20% error	2	0.210	0.239	0.269	0.291	0.297	0.301	0.314
AR(1), 50% error	0.5	0.257	0.315	0.347	0.378	0.381	0.385	0.400
AR(1), 50% error	1	0.243	0.275	0.305	0.326	0.332	0.340	0.352
AR(1), 50% error	2	0.230	0.247	0.259	0.265	0.261	0.258	0.257

Under CS, more switches never help, and once carryover is substantial they hurt: each switch puts an off-drug visit soon after a discontinuation, where moderation survives. Under AR(1), more switches help, because each brings on and off visits closer together, but the gain shrinks as the half-life lengthens and as measurement error grows; at a two-week half-life with 50% measurement error it has vanished (0.230 to 0.265, with the best at four switches).

5. Time between visits

With the number of visits fixed at eight, the spacing d between visits trades two effects against each other: on and off visits close together share more of the slowly varying noise, which the contrast cancels (correlation ρ^d); off-drug visits far from discontinuation retain less moderation ($2^{-d/t_{1/2}}$). Figure 1 shows power against spacing for three arrangements (verified, closed form).

Figure 1. Power against the time between visits for three blinded counterbalanced arrangements, by response half-life (light to dark, 0.25 to 4 weeks), under CS and AR(1) with 0%, 20% and 50% measurement error. Eight visits, so the trial lasts eight times the spacing. Points mark the spacing of highest power on the 0.25-6 week grid.

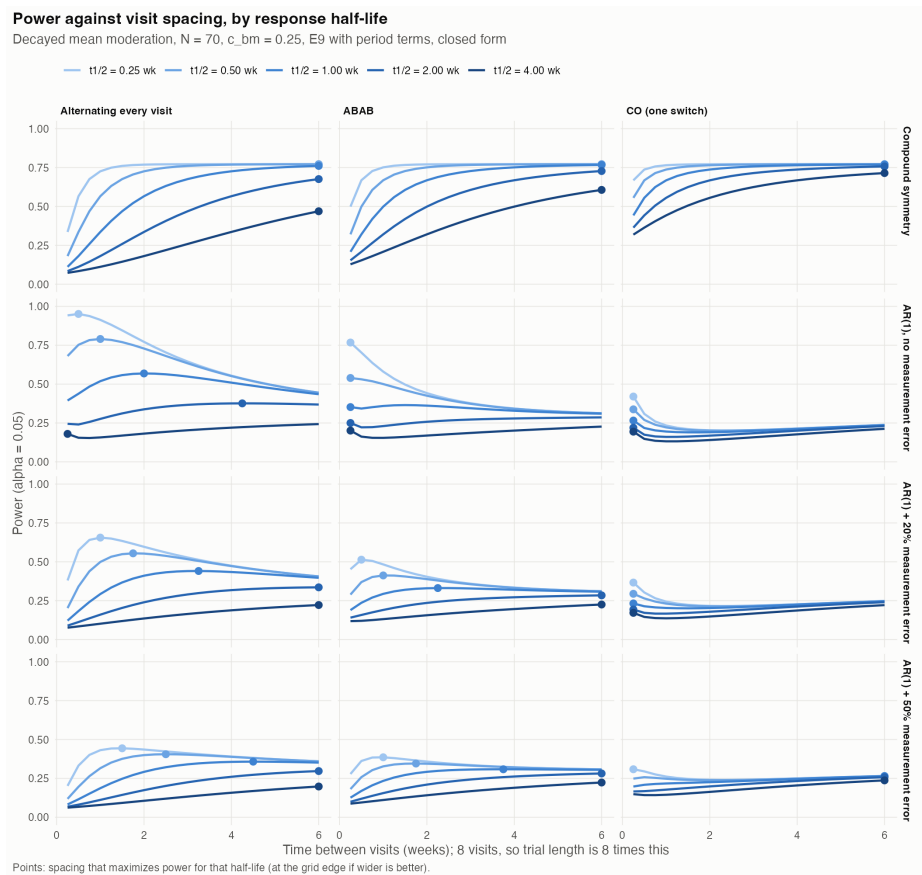


Figure 1: Power against visit spacing

Under CS there is no interior optimum. Correlation does not depend on spacing, so wider is always better, up to a plateau. The practical choice is the shortest spacing within 95% of the best power: for the alternating design 1.25, 2.25, 4.0 and 5.25 weeks at half-lives of 0.25, 0.5, 1 and 2 weeks, four to five half-lives at the shorter half-lives and 2.6 at two weeks; for ABAB 1.0, 1.75, 3.25 and 4.75 weeks, about 2.4 to 4 half-lives; for the crossover 0.5, 1.0, 2.0 and 3.25 weeks, about two, because its placebo-first sequence carries no carryover and its off-drug blocks extend well beyond the discontinuation.

Under AR(1) there is an interior optimum. For the alternating design (verified):

Measurement error	Optimal spacing at $t_{1/2}$ = 0.25 / 0.5 / 1 / 2	As multiple of $t_{1/2}$
	weeks	
none	0.5 / 1.0 / 2.0 / 4.25	about 2
20%	1.0 / 1.75 / 3.25 / beyond 6	about 3-4
50%	1.5 / 2.5 / 4.5 / beyond 6	about 4.5-6

Without measurement error the optimum has a closed form in the large-sample limit. Each off-drug visit is one spacing after a discontinuation, and the AR(1) time part of an alternating contrast is about $\frac{1}{2}(1 - \rho^d)/(1 + \rho^d)$, so information is proportional to

$$f(d) = (1 - 2^{-d/t_{1/2}})^2 \frac{1 + \rho^d}{1 - \rho^d},$$

whose maximum, 0.45, 0.91, 1.88 and 4.34 weeks at the four half-lives, agrees with the exact grid optimum (verified). The more of the noise is measurement error, the less close spacing buys, and the further the optimum moves out. ABAB optima are shorter (0.5, 1.0 and 2.25 weeks at half-lives of 0.25, 0.5 and 1 with 20% error), because the second visit of each off-drug block is already two spacings after discontinuation. The best attainable power falls as the half-life lengthens: a drug with long carryover is intrinsically harder to study within patients.

6. Exhaustive design search

Space. Every on/off pattern of the eight visits with two to six on-drug visits, paired with its complement as the second sequence; a pattern and its complement are the same design, leaving 119 designs. Each was evaluated at spacings of

0.5, 1, 1.5, 2, 2.5, 3 and 4 weeks, under the three structures and three half-lives: 7,497 design evaluations.

Best design in each condition (pattern of the first sequence; verified):

Structure	$t_{1/2}$	Best pattern	Switches	Spacing (weeks)	Power
CS	0.5	11110000 (CO)	1	4.0	0.765
CS	1	11110000 (CO)	1	4.0	0.759
CS	2	11110000 (CO)	1	4.0	0.731
AR(1), 20% error	0.5	10101010 (alternating)	7	2.0	0.547
AR(1), 20% error	1	10101010	7	3.0	0.435
AR(1), 20% error	2	10101010	7	4.0	0.314
AR(1), 50% error	0.5	10101010	7	2.5	0.400
AR(1), 50% error	1	10101010	7	4.0	0.352
AR(1), 50% error	2	10011001	4	4.0	0.265

Under CS the optimum lies at the edge of the spacing grid, consistent with Section 5: wider would be better still, though by little (the CS plateau). Under AR(1) the optimum is interior for the shorter half-lives and moves to the edge as the half-life grows.

The landscape (Figure 2). Under CS most balanced designs cluster near the best, and the lower bands are the designs with unequal numbers of on and off visits, as Section 3 predicts: at a half-life of 0.5 weeks, power is 0.763-0.765 with four on-drug visits, 0.737-0.739 with three or five, and 0.643-0.645 with two or six (verified). Under AR(1) power rises steadily with the number of switches.

Figure 2. Every one of the 119 designs at its best spacing, against the number of switches in its first sequence, for each structure (rows) and half-life (columns). Dark open symbols: the crossover, ABAB and alternating reference designs. Orange: the best design in each condition.

Robustness. The correlation structure and the response half-life of a real drug and outcome are uncertain, so a design that performs well across conditions is more useful than one that is optimal in one. The design-spacing pairs with the

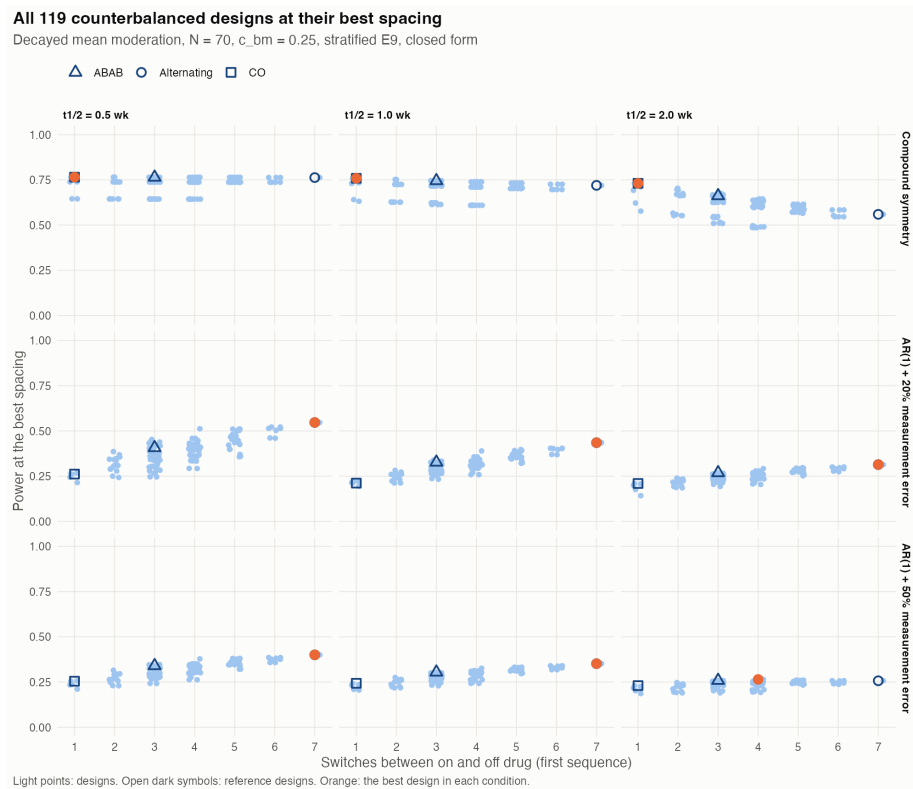


Figure 2: All designs at their best spacing

highest minimum, over the nine conditions, of power relative to that condition's best (verified):

Pattern (first sequence)	Spacing (weeks)	Worst relative power	Mean relative power
10101001 (tied with its mirror image 10010101)	4	0.78	0.92
10100101	4	0.77	0.91
10101010 (alternating)	4	0.77	0.94
10011001	3	0.76	0.88

The robust designs are alternating or nearly so, at wide spacing. They give up some power under CS with long carryover, where frequent switching is penalized, in exchange for most of the AR(1) gain.

7. Open-label components

Hendrickson et al.'s Hybrid and OL+BDC designs begin with open-label visits, often needed for dose titration. In the data-generating model, expectancy e_t (1 at open-label visits, 0.5 at blinded ones) scales the placebo (expectancy) factor's mean and standard deviation. Because that factor is unrelated to the biomarker in the model, the higher placebo response during open-label visits does not bias the interaction slope; it changes only the noise. The search was repeated with $k = 0$ to 4 open-label on-drug visits for every patient, followed by counterbalanced blinded sequences over the remaining $8 - k$ visits, and two analyses: the open-label visits included in the on/off contrast, as E9 and the published analysis do, or left out of it. Best power over patterns and spacings (verified):

Structure	$t_{1/2}$	$k = 0$	1, in / out	2, in / out	3, in / out	4, in / out
CS	0.5	0.765	0.722 / 0.701	0.679 / 0.644	0.605 / 0.551	0.547 / 0.479
CS	1	0.759	0.707 / 0.686	0.661 / 0.627	0.583 / 0.530	0.521 / 0.456
CS	2	0.731	0.638 / 0.616	0.582 / 0.549	0.491 / 0.443	0.421 / 0.366
AR(1), 20% error	0.5	0.547	0.463 / 0.487	0.355 / 0.422	0.280 / 0.361	0.237 / 0.291

Structure	$t_{1/2}$	$k = 0$	1, in / out	2, in / out	3, in / out	4, in / out
AR(1), 20% error	1	0.435	0.370 / 0.374	0.306 / 0.328	0.251 / 0.274	0.215 / 0.228
AR(1), 20% error	2	0.314	0.257 / 0.258	0.217 / 0.228	0.181 / 0.192	0.160 / 0.163
AR(1), 50% error	0.5	0.400	0.358 / 0.352	0.311 / 0.311	0.262 / 0.261	0.231 / 0.220
AR(1), 50% error	1	0.352	0.314 / 0.305	0.278 / 0.271	0.238 / 0.227	0.210 / 0.194
AR(1), 50% error	2	0.265	0.228 / 0.217	0.208 / 0.196	0.179 / 0.165	0.161 / 0.144

- **Each open-label visit that replaces a blinded visit costs power,** under every structure and half-life: with the better of the two analyses, two open-label visits cost 0.09 to 0.15 under CS and 0.09 to 0.13 under AR(1) with 20% error. Part of the cost is the lost blinded visit; part is the open-label visits' noisier, confounded contribution.
- **Whether to include them in the contrast depends on the structure.** Under CS including them is better, plausibly because they add on-drug information that outweighs their expectancy noise (inferred). Under AR(1) with 20% measurement error leaving them out is better, plausibly because they come early in the trial, far from the blinded off-drug visits, and so add noise that the contrast cannot cancel (inferred). With 50% measurement error the two are close.

This comparison holds the total number of visits fixed. If an open-label titration phase is needed clinically, it can be added before the blinded visits rather than in place of them; left out of the contrast, it then costs no power, and included in it, its effect follows the pattern above. That variant was not evaluated.

8. Monte Carlo checks

The stratified E9 statistic was simulated directly, 5000 trials of 70 patients per design, for the best design in each of the nine conditions, for the crossover, ABAB and alternating designs at 2.5-week spacing in every condition, and for the best two-visit open-label designs at a one-week half-life under both analyses: 41 designs. The draws come from the full joint distribution, including the AR(1)

structures with measurement error, which no earlier check covered. Every design agrees with the closed form within two Monte Carlo standard errors (largest deviation 1.97, mean +0.35; verified). The closed form is, if anything, slightly conservative.

9. How to approach an optimal N-of-1 design

The analysis suggests the following procedure; the principles are inferred from the closed form and confirmed for the designs simulated.

1. **State the two uncertain inputs.** The response half-life $t_{1/2}$ (docs/38: the persistence of benefit after discontinuation, not the drug's plasma half-life) and the within-patient correlation structure, in particular how fast correlation decays and how much of the variability is measurement error. Both can be estimated from pilot or historical data, and the design should be chosen to perform well across their plausible ranges.
2. **Blind the contrast, and balance it.** Equal numbers of on-drug and off-drug assessments per patient, with expectancy the same at on and off assessments. If open-label titration is necessary, add it before the blinded phase rather than in place of blinded assessments.
3. **Interleave if correlation decays.** Under AR(1)-like correlation, alternate frequently so that every on-drug assessment has an off-drug assessment nearby. Under CS-like correlation, arrangement does not matter and fewer, longer blocks are better under carryover.
4. **Space assessments against the response half-life.** Off-drug assessments should fall several half-lives after discontinuation: up to four or five under CS for alternating designs, two to six under AR(1) depending on the measurement-error share, fewer for long blocks in which most off-drug assessments are late.
5. **Counterbalance sequences and analyze with period terms.** Sequence and period differences otherwise enter the residual and cost power.
6. **Weight the contrast by the covariance.** E9 weights all on-drug and all off-drug visits equally, which is not optimal under AR(1); the best linear contrast is proportional to C^{-1} times the moderation pattern, which a correctly specified mixed model approximates.
7. **Search the design space with the closed form.** The evaluation of one design is instantaneous, so the full space of sequences and spacings can be searched for the conditions of a given trial, and the shortlisted designs confirmed by simulation, as in Sections 6 and 8.

10. Limitations

- **Fixed visit count and sample size.** Eight post-baseline visits, $N = 70$, two sequences of equal size. Trial length therefore varies with spacing; a

fixed trial length would trade visits against spacing, a different question. Unequal spacing within a trial was not searched.

- **The data-generating rule.** Decayed mean moderation with anchored decay. Under the covariance constructs, high-information designs have lower c_{bm} ceilings and could not represent the same effect sizes; under proportional moderation (docs/38), on-drug moderation would also build up during titration.
- **The correlation structures.** CS and separable AR(1) with $\rho = 0.8$ per week and 0%, 20% or 50% measurement error. Which describes real symptom data is an empirical question that decides the ranking of designs.
- **The analysis.** Stratified E9 with equal weights. A mixed model with the correct covariance would gain somewhat, most under AR(1).
- **Expectancy.** The placebo factor is unrelated to the biomarker in the model. If expectancy effects were themselves moderated by the biomarker, open-label phases would bias the interaction, not only add noise.
- **Not all designs simulated.** The search is closed form; 41 designs were simulated, all agreeing.

11. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/02-design-characteristics.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/03-optimal-spacing.R
Rscript analysis/scripts/quick-sim/carryover-closed-form/04-design-search.R \
--mc-reps 5000 --cores 6
```

Run from the repository root. All three scripts evaluate the definitions of 01-e9-closed-form.R (docs/37). Outputs are written to analysis/data/quick-sim/carryover-closed-form/ (design structure and spacing tables) and its design-search/ subdirectory (the full search, best designs, robustness ranking, open-label search and Monte Carlo check), and the figures to docs/figures/39-fig1-optimal-spacing.png and docs/figures/39-fig2-design-landscape.png.

12. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. Code: github.com/rchendrickson/pmsimstats, commit 58b32a9.
2. pmsimstats team. docs/37-carryover-power-closed-form.md (closed-form moments, the factor-by-time decomposition and the validation of the framework).
3. pmsimstats team. docs/38-pharmacodynamic-carryover-model.md (the response half-life and proportional moderation).

4. pmsimstats team. docs/36-hendrickson-pd-and-carryover.md.